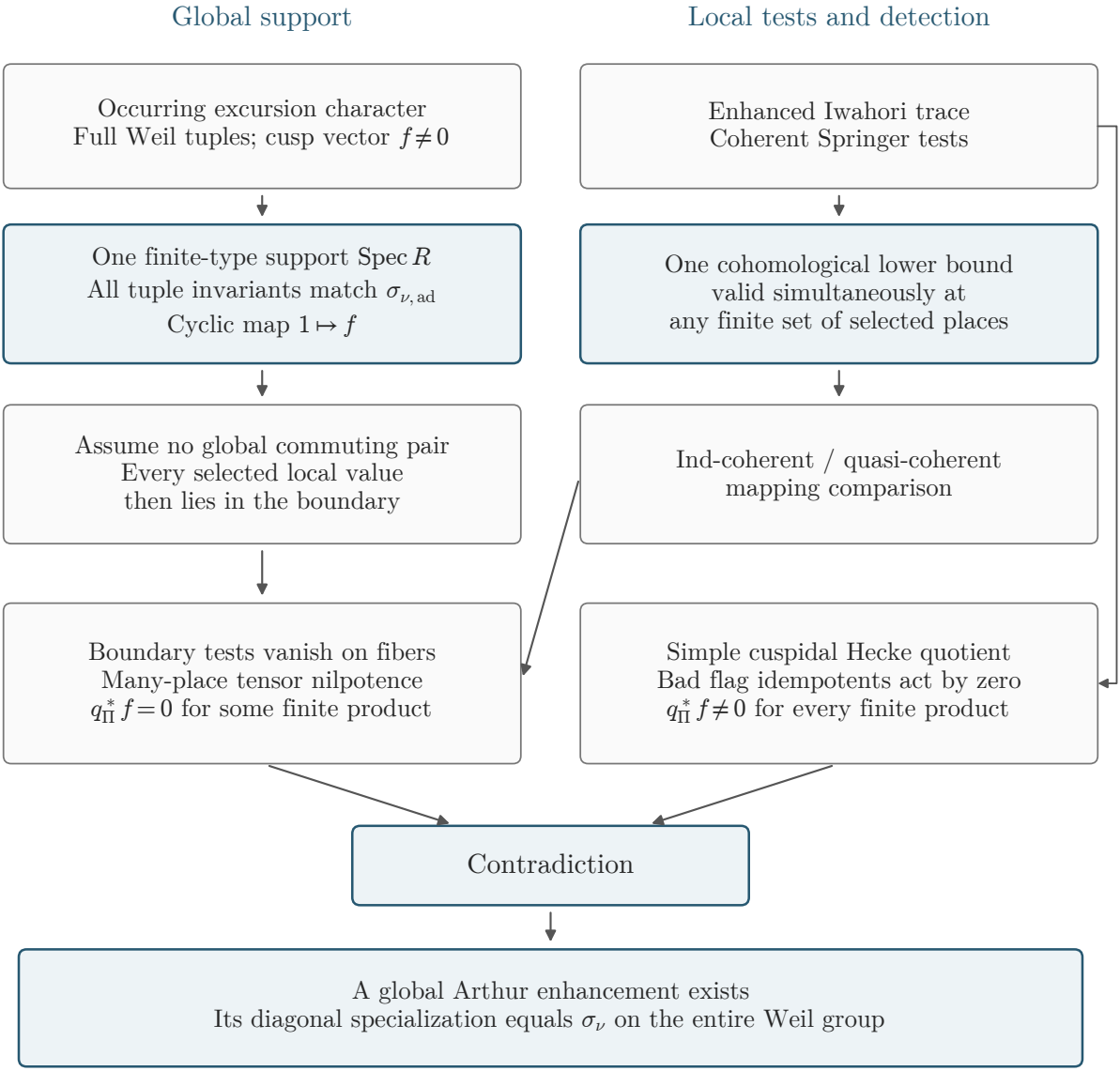
Global support
Local tests and detection
Occurring excursion character
Full Weil tuples; cusp vector $f \neq 0$
Enhanced Iwahori trace
Coherent Springer tests
One finite-type support $\operatorname{Spec} R$
All tuple invariants match $\sigma_{\nu,\mathrm{ad}}$
Cyclic map $1 \mapsto f$
One cohomological lower bound
valid simultaneously at
any finite set of selected places
Assume no global commuting pair
Every selected local value
then lies in the boundary
Ind-coherent / quasi-coherent
mapping comparison
Boundary tests vanish on fibers
Many-place tensor nilpotence
$q_{\Pi}^{*} f = 0$ for some finite product
Simple cuspidal Hecke quotient
Bad flag idempotents act by zero
$q_{\Pi}^{*} f \neq 0$ for every finite product
Contradiction
A global Arthur enhancement exists
Its diagonal specialization equals σ_{ν} on the entire Weil group